Acknowledgments:
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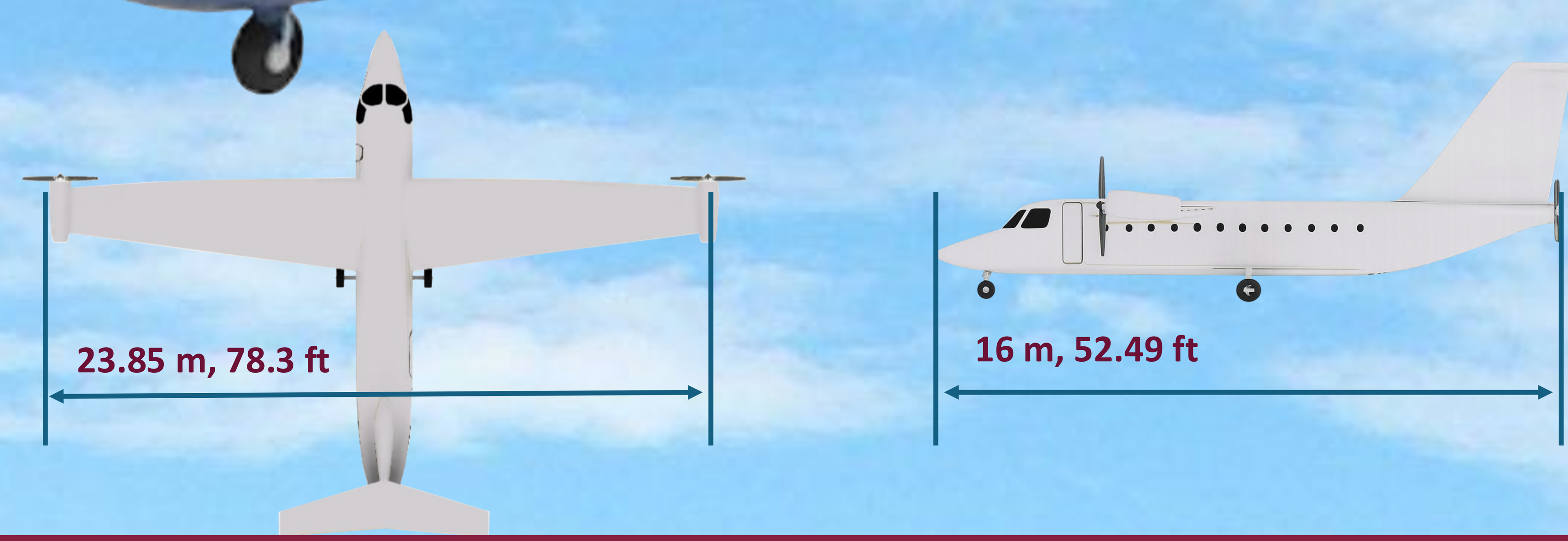
ECHO

Electrified Commuter Hybrid Operator



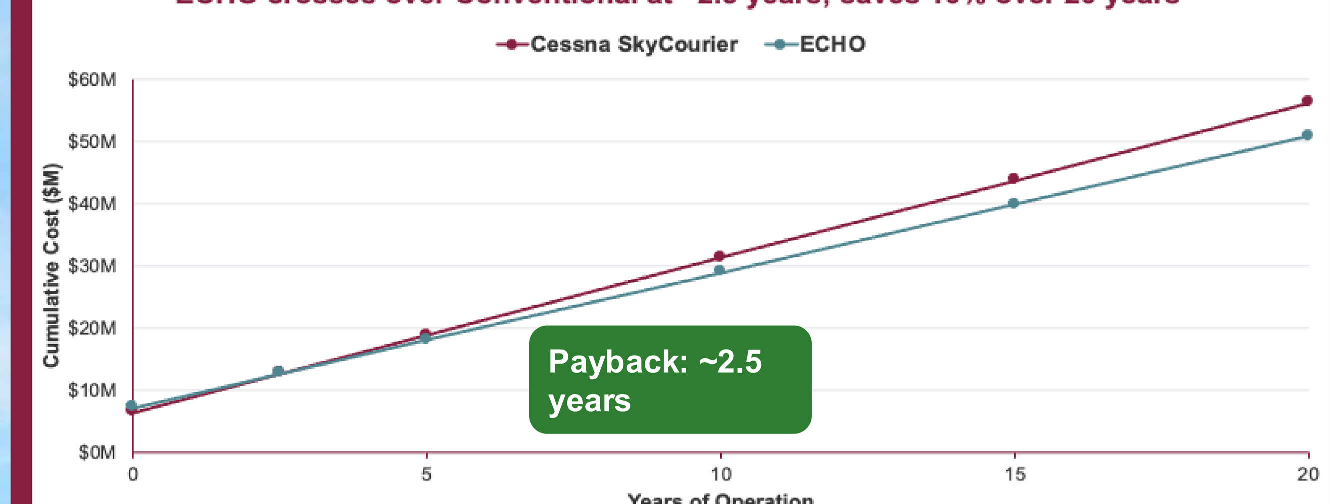
Executive Summary

21%	Lower operating cost per available seat-mile (CASM) vs. SkyCourier
19	Passenger FAR 23 hybrid aircraft
$115B	Total addressable market
6 x	More airports accessible than regional jets.

Cost Modeling

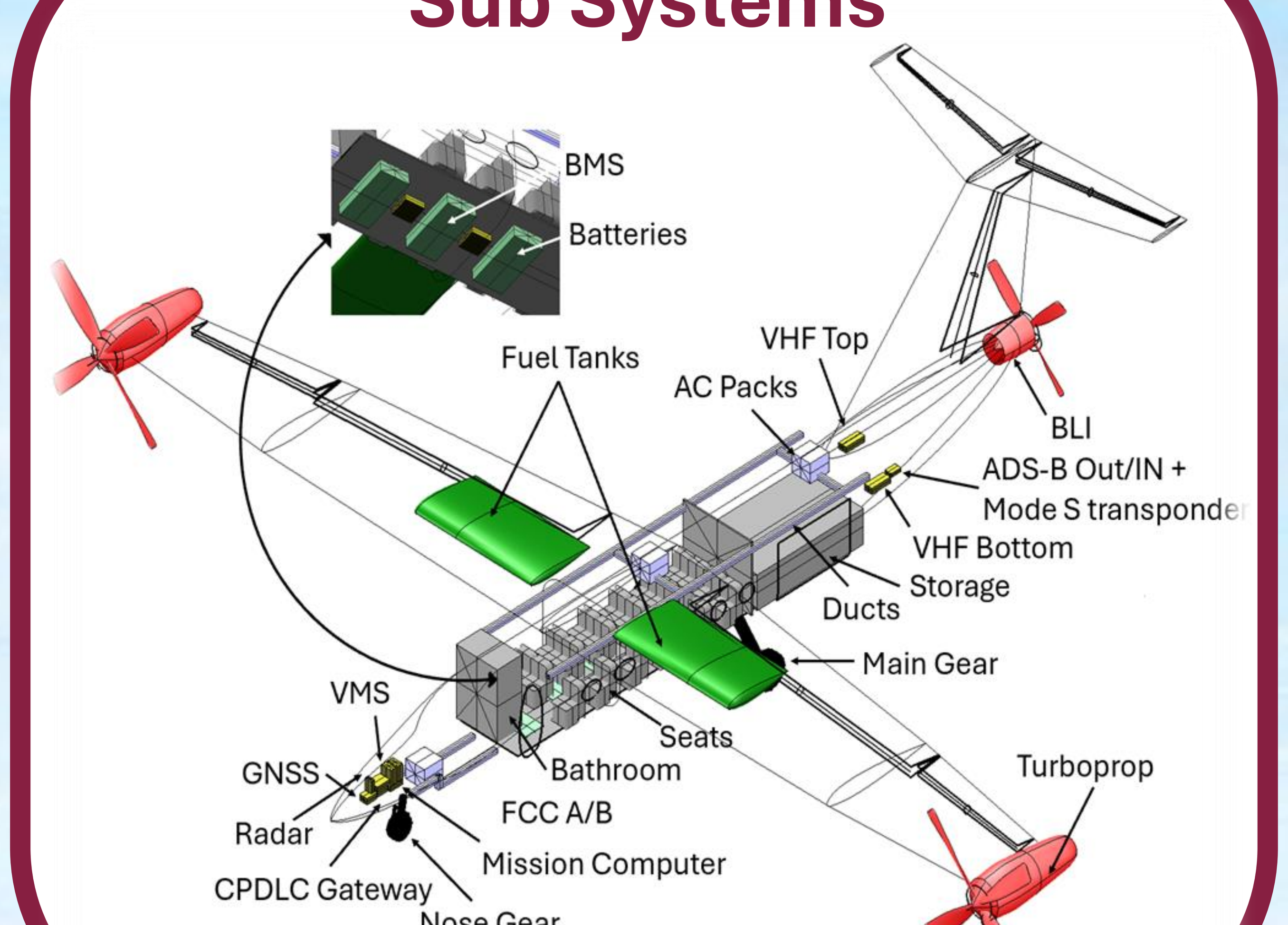
$5.4M 20-yr LCC savings vs. comparator

ECHO crosses over Conventional at ~2.5 years; saves 10% over 20 years

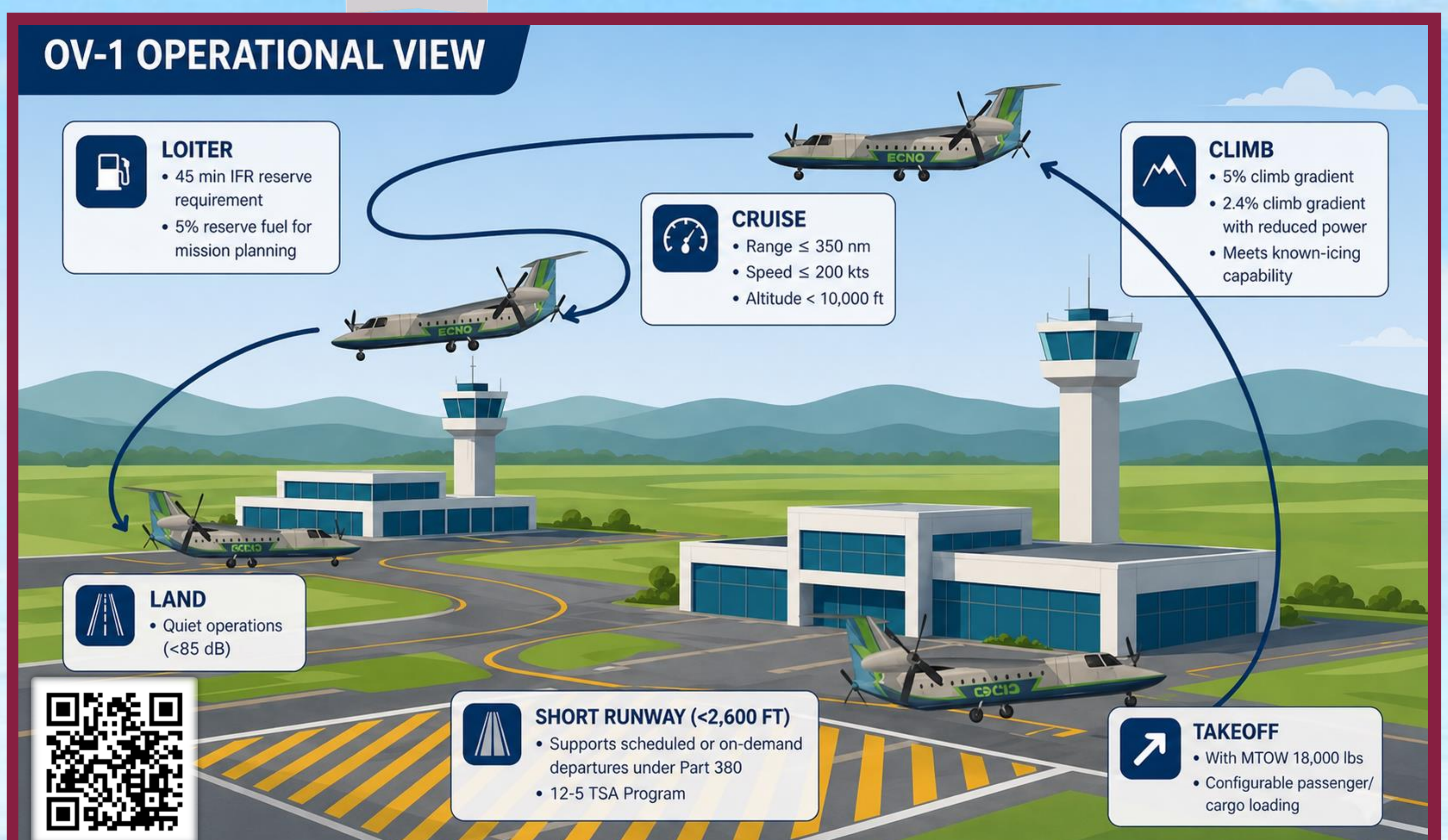


- ECHO's hybrid architecture trades acquisition capex for operating-cost reduction over the aircraft's service life.
- LCC model validated against Cessna SkyCourier within 5%

Sub Systems



OV-1 OPERATIONAL VIEW



OEI Scenario

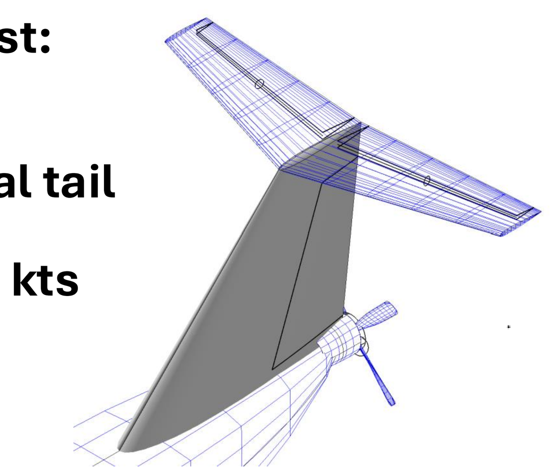
Massive asymmetric thrust:

Yaw trimmed by both tail propeller and large vertical tail

Controllable even with 30 kts wind.

V_{mc} = **81 kts** (FAR 23 min.)

Y_{OEI} = **2.40%** (FAR 23 min.)



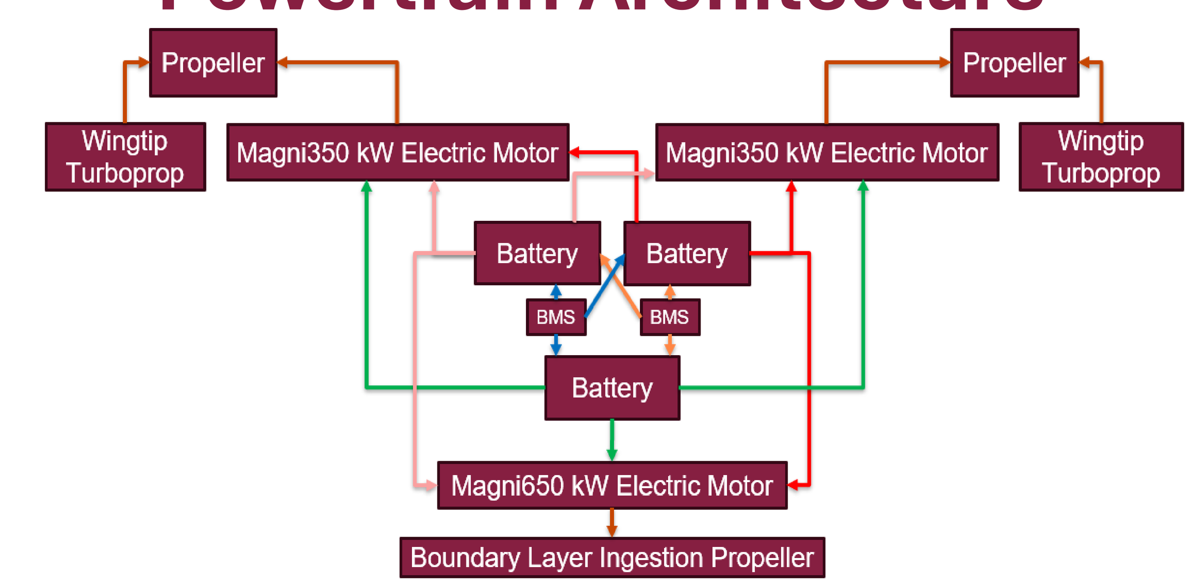
Takeoff and Landing

Takeoff Distances	
Dry Concrete	1,930
Wet Concrete	2,012
Icy Concrete	2,199
Landing Distances	
Dry Concrete	2,275
Wet Concrete	2,403
Icy Concrete	2,577

E.C.H.O. was designed to takeoff and land under 2,600 feet and is capable of this on dry, wet, and icy runway conditions.

E.C.H.O. has a balanced field length constrained to 2,600 feet to ensure safe takeoff.

Powertrain Architecture



MagniX motors are powered by magniDrive 100s. These are bi-directional inverters, allowing battery recharge mid-flight using the torque generated by turboprop operation during cruise and descent.

Subscale Demonstrator

Validated **15%** propulsive efficiency increase and controllability during flight in this configuration